Home assignment 2 – Analysis of a bent beam structure

	FIGURE	CODE2	CODE3	CODE4
Task code:	1	3	1	3

NEPTUN code	NAME
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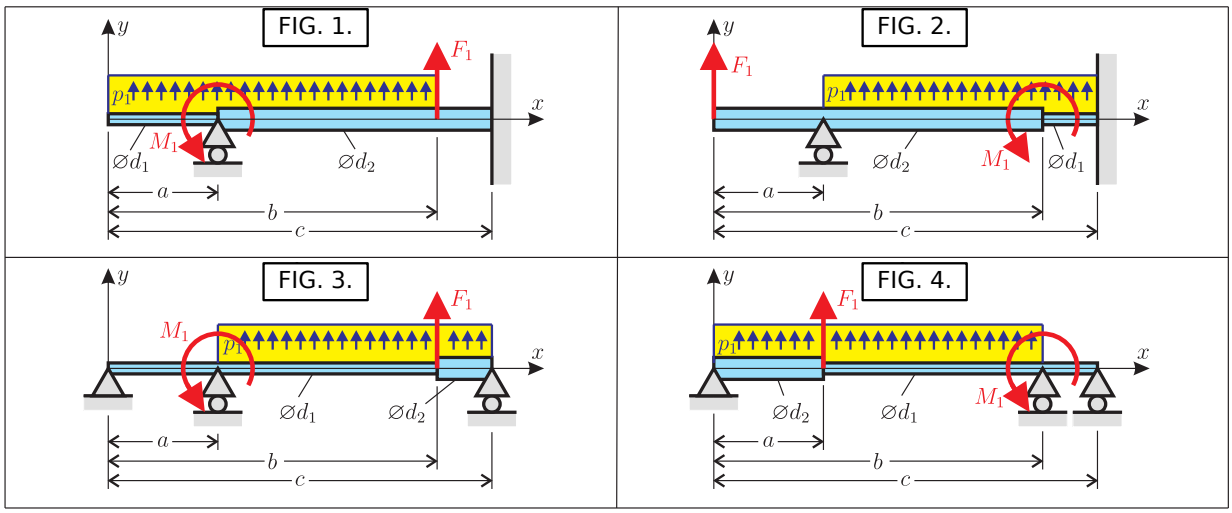


The structures in the figures are subjected to a uniform distributed load p_1, a concentrated force F_1 and a concentrated torque M_1. The structures consist of two sections with circular cross-sections of different diameters ($d_1 = d$ and $d_2 = 2d$). The material of the structures is linear elastic, isotropic, homogeneous. The d_1 diameter section has an elastic modulus $4E$, the d_2 diameter section has an elastic modulus E. The density of the material is ρ. Based on the personal task code, the data of the task can be found in the following table:

	CODE2		CODE3			CODE4			
	E [GPa]	d [mm]	a [mm]	b [mm]	c [mm]	p_1 [N/m]	F_1 [kN]	M_1 [kNm]	ρ [kg/m^3]
CODE=1	50	23	220	540	730	2500	4	0.6	2500
CODE=2	40	27	230	460	610	−2500	−3	−0.75	2500
CODE=3	30	31	430	550	890	3000	2	0.9	2500
CODE=4	20	35	330	440	680	−3000	−1	−1.1	2500

Build the finite element model by using 1D BEAM elements! Calculate all tasks both

- **I.** by using Matlab/Octave or other program code
- **II.** by using Ansys (only tasks 1,2,5).

1. Determine the translation and rotation of the nodes!
2. Calculate the reaction forces!
3. Plot the following functions: deflection ($u_y(x)$), rotation ($\varphi_z(x)$), bending moment ($M_b(x)$)!
4. Determine the value of the deflection function $u_y(x)$ at the middle points of the elements!
5. Determine the first three natural frequencies of the unloaded structure! (Use consistent mass matrices.)

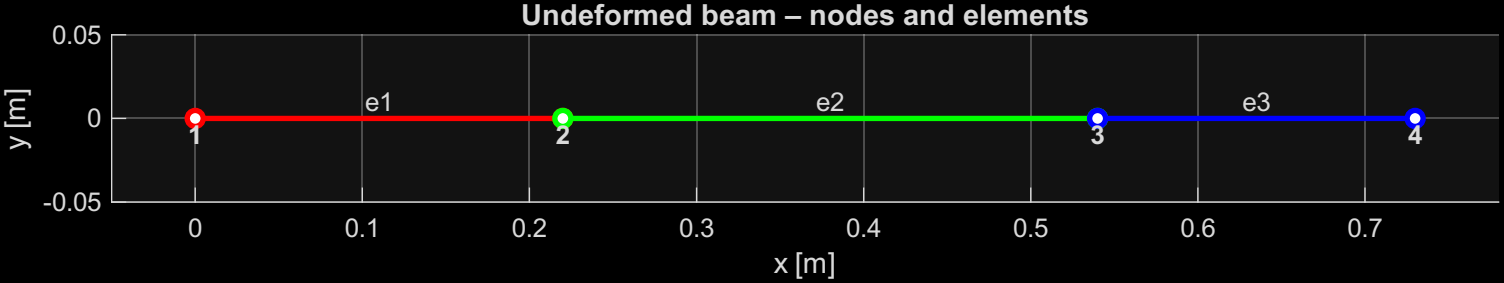
Fill the tables!

DISPLACEMENTS		
node	u_y [mm]	φ_z [deg]
1.	-1.091	0.270
2.	0	0.326
3.	0.319	-0.121
4.	0	0

REACTION FORCES		
csp.	F [kN]	M [kNm]
1.	0	0
2.	0.871	0
3.	0	0
4.	-4.491	0.669

DEFLECTION	
x	$u_y(x)$ [mm]
$a/2$	-0.5724
$(a+b)/2$	0.4722
$(b+c)/2$	0.1093

NATURAL FREQ.	
frequency [Hz]	
f_1	418.166
f_2	685.401
f_3	2012.553


Undeformed beam – nodes and elements
0.05
0
-0.05
y [m]
e1
e2
e3
1
2
3
4
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
x [m]

Given Data

The beam structure consists of three beam elements with two different circular cross-sections.

$$E = 30\,\text{GPa}, \qquad a = 220\,\text{mm}, \qquad b = 540\,\text{mm}, \qquad c = 730\,\text{mm}, \qquad d = 31\,\text{mm}.$$

$$p_1 = 3000\,\text{N/m}, \qquad F_1 = 2\,\text{kN}, \qquad M_1 = 0.9\,\text{kNm}, \qquad \rho = 2500\,\text{kg/m}^3.$$

Cross-section diameters:

$$d_1 = d, \qquad d_2 = 2d = 62\,\text{mm}.$$

Second moments of area:

$$I_1 = \frac{\pi d_1^4}{64}, \qquad I_2 = \frac{\pi d_2^4}{64}.$$

Element lengths:

$$L_1 = a, \qquad L_2 = b - a, \qquad L_3 = c - b.$$

DOF Numbering and Index Sets

Each node has:

$$[w_i,\ \varphi_i].$$

Global DOF vector:

$$\mathbf{U} = [w_1, \varphi_1,\ w_2, \varphi_2,\ w_3, \varphi_3,\ w_4, \varphi_4]^T.$$

Element DOF Indices

$$ie_1 = [1,\ 2,\ 3,\ 4], \qquad ie_2 = [3,\ 4,\ 5,\ 6], \qquad ie_3 = [5,\ 6,\ 7,\ 8].$$

Boundary Conditions

For FIGURE 1 (left free, middle roller, right built-in):

$$i_{\text{fixed}} = [3,\ 7,\ 8], \qquad i_{\text{free}} = [1,\ 2,\ 4,\ 5,\ 6].$$

Finite Element Model and System Equations

In this task the beam is modelled by 3 Euler–Bernoulli beam elements and 4 nodes. Each element has the local degrees of freedom

$$u_e = [\, w_i,\ \varphi_i,\ w_j,\ \varphi_j \,]^T.$$

1. Element stiffness matrices

For a beam element of length L and flexural rigidity EI the local stiffness matrix is

$$\mathbf{k}_e = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}.$$

In this homework:

$$\begin{aligned} e_1 &: (1{-}2), \quad L_1 = a, \qquad E_1 = 4E, \\ e_2 &: (2{-}3), \quad L_2 = b - a, \quad E_2 = E, \\ e_3 &: (3{-}4), \quad L_3 = c - b, \quad E_3 = E. \end{aligned}$$

So we compute

$$\mathbf{k}_1 = \mathbf{k}_e(L_1, EI_1), \quad \mathbf{k}_2 = \mathbf{k}_e(L_2, EI_2), \quad \mathbf{k}_3 = \mathbf{k}_e(L_3, EI_3),$$

and assemble them into the global 8×8 stiffness matrix $\mathbf{K}$ using

$$ie_1 = [1, 2, 3, 4], \quad ie_2 = [3, 4, 5, 6], \quad ie_3 = [5, 6, 7, 8].$$

In practice this means that each $\mathbf{k}_e$ is added into the correct rows and columns of $\mathbf{K}$ according to its index vector ie_e.

2. Load vector

The total nodal load vector is the sum of:

- the equivalent nodal loads of the distributed load p_1,
- the concentrated torque M_1 at node 2,
- the concentrated force F_1 at node 3.

For a uniform load p_1 on a beam element of length L (positive upwards) the equivalent nodal loads are

$$f_q^{(e)} = p_1 L \begin{bmatrix} \frac{1}{2} \\ \frac{L}{12} \\ \frac{1}{2} \\ -\frac{L}{12} \end{bmatrix}.$$

In this problem the distributed load acts on elements 1 and 2, so we form

$$f_q^{(1)} = p_1 L_1 \begin{bmatrix} \frac{1}{2} \\ \frac{L_1}{12} \\ \frac{1}{2} \\ -\frac{L_1}{12} \end{bmatrix}, \qquad f_q^{(2)} = p_1 L_2 \begin{bmatrix} \frac{1}{2} \\ \frac{L_2}{12} \\ \frac{1}{2} \\ -\frac{L_2}{12} \end{bmatrix},$$

and add their components into the global load vector F_a at DOFs ie_1 and ie_2 respectively.

The concentrated loads are added directly to the corresponding DOFs:

$$M_1 \text{ at } \varphi_2 \Rightarrow F_4 = F_4 + M_1, \qquad F_1 \text{ at } w_3 \Rightarrow F_5 = F_5 + F_1.$$

Complete Global Load Vector

The total load vector is built as the sum of the individual contributions:

$$F_a = F_{q1} + F_{q2} + F_{M_1} + F_{F_1}.$$

UDL on element 1 (nodes 1–2):

$$F_{q1} = p_1 L_1 \begin{bmatrix} \frac{1}{2} \\ \frac{L_1}{12} \\ \frac{1}{2} \\ -\frac{L_1}{12} \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

UDL on element 2 (nodes 2–3):

$$F_{q2} = p_1 L_2 \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2} \\ \frac{L_2}{12} \\ \frac{1}{2} \\ -\frac{L_2}{12} \\ 0 \\ 0 \end{bmatrix}.$$

Concentrated moment at node 2:

$$F_{M_1} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ M_1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Concentrated force at node 3:

$$F_{F_1} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ F_1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Using the global DOF order

$$[w_1,\ \varphi_1,\ w_2,\ \varphi_2,\ w_3,\ \varphi_3,\ w_4,\ \varphi_4],$$

the complete assembled vector is:

$$F_a = \begin{bmatrix} p_1 L_1 \dfrac{1}{2} \\ p_1 L_1 \dfrac{L_1}{12} \\ p_1 L_1 \dfrac{1}{2} \ + \ p_1 L_2 \dfrac{1}{2} \\ -p_1 L_1 \dfrac{L_1}{12} \ + \ p_1 L_2 \dfrac{L_2}{12} \ + \ M_1 \\ p_1 L_2 \dfrac{1}{2} \ + \ F_1 \\ -p_1 L_2 \dfrac{L_2}{12} \\ 0 \\ 0 \end{bmatrix}.$$

3.1 Condensation of the system

Split fixed and free DOFs:

$$i_{\text{fixed}} = [3, 7, 8], \qquad i_{\text{free}} = [1, 2, 4, 5, 6].$$

Keep only the rows and columns of the free DOFs:

$$\boxed{K_c\, U_c = F_c}$$

3.2 Solve for condensed displacements

$$\boxed{U_c = K_c^{-1}\, F_c}$$

3.3 Expand back to full displacement vector

Insert zeros at the fixed DOFs, and put the values of U_c at the free DOFs:

$$\boxed{U = \text{expand}\big(U_c;\ i_{\text{fixed}}, i_{\text{free}}\big)}$$

This gives the nodal deflections and rotations

$$U = [\, w_1,\ \varphi_1,\ w_2,\ \varphi_2,\ w_3,\ \varphi_3,\ w_4,\ \varphi_4 \,]^T.$$

3.4 Total nodal forces

$$\boxed{F = K\, U}$$

3.5 Reaction forces at supports

Using equilibrium $F_r + F_a = KU$,

$$\boxed{F_r = K\, U - F_a}$$

The components of F_r at the fixed DOFs (i_{fixed}) are the support reactions.

3.6 Element end forces and bending stresses

For each element e we extract its nodal DOFs U_e and compute

$$F_e = K_e\, U_e$$

(end shear forces and end bending moments).

The bending moment distribution $M(x)$ inside the element is obtained from F_e, and the normal stress at the outer fibre is

$$\boxed{\sigma(x) = \frac{M(x)}{I}\, z,}$$

with $z = d_1/2$ or $d_2/2$ depending on the section.

3.7 Mass matrix and natural frequencies

For each beam element of length L, area A and density ρ, the consistent mass matrix is

$$M_e = \rho A L \begin{bmatrix} \frac{156}{420} & \frac{22L}{420} & \frac{54}{420} & -\frac{13L}{420} \\ \frac{22L}{420} & \frac{4L^2}{420} & \frac{13L}{420} & -\frac{3L^2}{420} \\ \frac{54}{420} & \frac{13L}{420} & \frac{156}{420} & -\frac{22L}{420} \\ -\frac{13L}{420} & -\frac{3L^2}{420} & -\frac{22L}{420} & \frac{4L^2}{420} \end{bmatrix}.$$

All M_e are assembled into the global mass matrix M using the same index vectors ie_1, ie_2, ie_3 as for the stiffness matrix.

The free–vibration problem is

$$\boxed{\left(K - \omega^2 M\right)\phi = 0}$$

and the natural circular frequencies ω_n follow from the eigenvalues. The natural frequencies in Hz are:

$$\boxed{f_n = \frac{\omega_n}{2\pi}}.$$

Summary of Calculation Steps

We first numbered all nodes and elements and assigned two degrees of freedom per node (the transverse displacement w and the rotation φ). Then we calculated the length and second moment of area of each beam element, and listed the fixed and free DOFs based on the supports:

$$i_{\text{fixed}} = [3, 7, 8], \qquad i_{\text{free}} = [1, 2, 4, 5, 6].$$

Using the Euler–Bernoulli beam element stiffness matrix K_e, we built the stiffness matrix of every element and assembled them into the global stiffness matrix K. Next, we constructed the global load vector by adding the equivalent nodal loads of the distributed load, and the concentrated moment and force. Then we applied the boundary conditions and condensed the system, keeping only the rows and columns of the free DOFs to obtain

$$K_c U_c = F_c.$$

We solved this system for the unknown free displacements with

$$U_c = K_c^{-1} F_c,$$

and expanded back to the full displacement vector U by inserting zeros at the fixed DOFs. From U we computed the total nodal force vector

$$F = KU,$$

and the reaction forces and moments from

$$F_r = KU - F_a.$$

For each element, we formed the end-force vector

$$F_e = K_e U_e,$$

and used it to obtain the internal shear forces, bending moments and normal stresses in the cross-sections. Finally, we assembled the consistent mass matrix from the element mass matrices and solved the eigenvalue problem

$$(K - \omega^2 M)\,\phi = 0$$

to determine the natural circular frequencies ω_n and the corresponding natural frequencies $f_n = \omega_n/(2\pi)$.

Results from MATLAB Analysis

Nodal Displacements

Node	w [mm]	φ [rad]
1	-1.0909	$+4.7141 \times 10^{-3}$
2	0.0000	$+5.6928 \times 10^{-3}$
3	$+0.3193$	-2.1197×10^{-3}
4	0.0000	0.0000

Reaction Forces and Moments

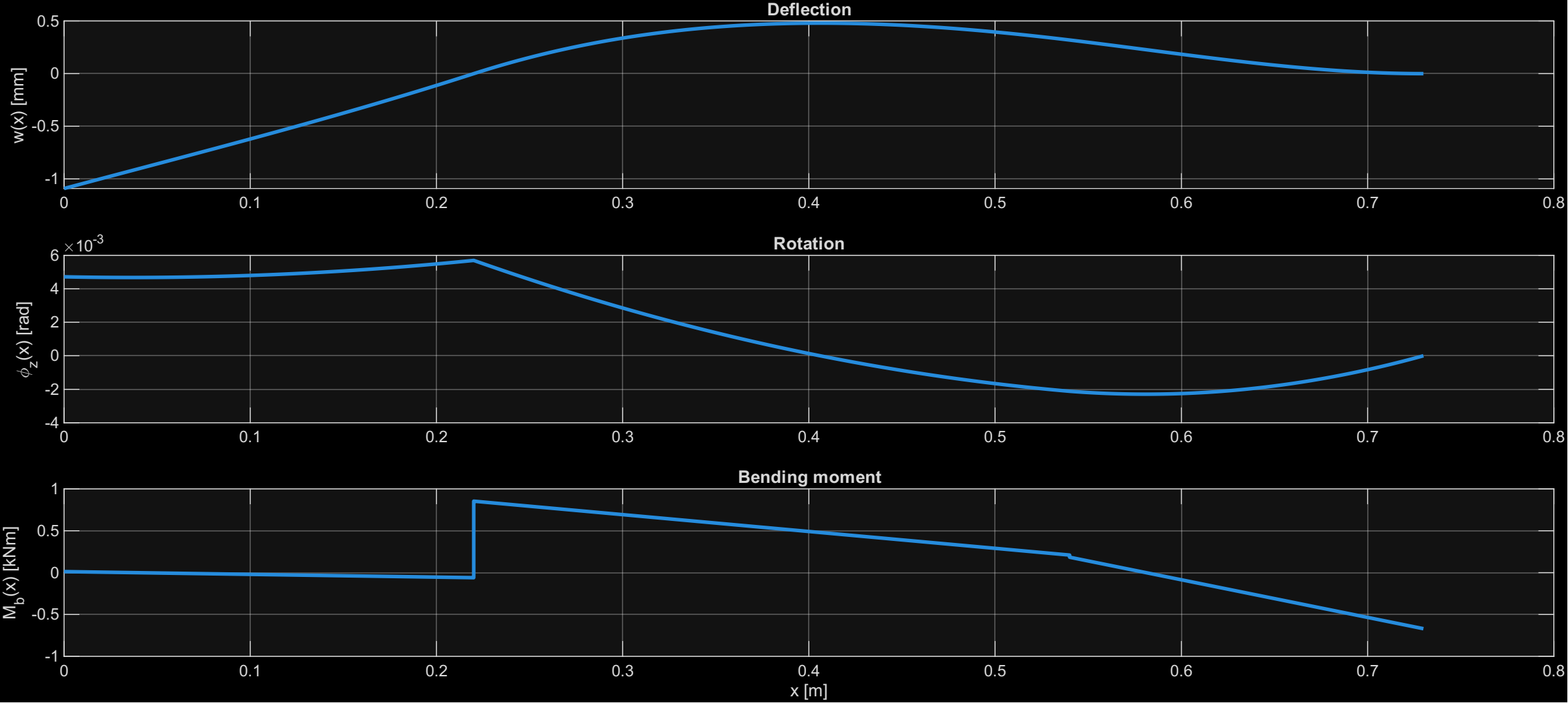
DOF	Node	Type	Reaction
3	2	w	$+0.871$ kN
7	4	w	-4.491 kN
8	4	φ	$+0.669$ kNm

Deflection at Element Midpoints

Position x	$u_y(x)$ [mm]
$a/2$	-0.5724
$(a+b)/2$	$+0.4722$
$(b+c)/2$	$+0.1093$

First Three Natural Frequencies

Mode	ω [rad/s]	f [Hz]
1	2627.414	418.166
2	4306.503	685.401
3	12645.243	2012.553


Deflection
w(x) [mm]
0.5
0
-0.5
-1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Rotation
×10^-3
6
4
2
0
-2
-4
φz(x) [rad]
Bending moment
1
0.5
0
-0.5
-1
Mb(x) [kNm]
x [m]

Results from ANSYS Analysis

Nodal Displacements

Node	w [mm]	φ [rad]
1	-1.1230	$+4.8702 \times 10^{-3}$
2	$+0.0000$	$+5.8509 \times 10^{-3}$
3	$+0.33938$	-2.0353×10^{-3}
4	$+0.0000$	$+0.0000$

Table 1: *
Nodal displacements w_i and rotations φ_i.

Reaction Forces and Moments

The fixed degrees of freedom are: DOF 3 at node 2 (w_2), and DOFs 7–8 at node 4 (w_4, φ_4). The corresponding reactions are:

Node	**DOF**	R_y [N]	M_z [N m]
2 (roller)	w_2	$+846.9540$	0
4 (built–in)	w_4	-4466.50	0
4 (built–in)	φ_4	0	$+656.930$

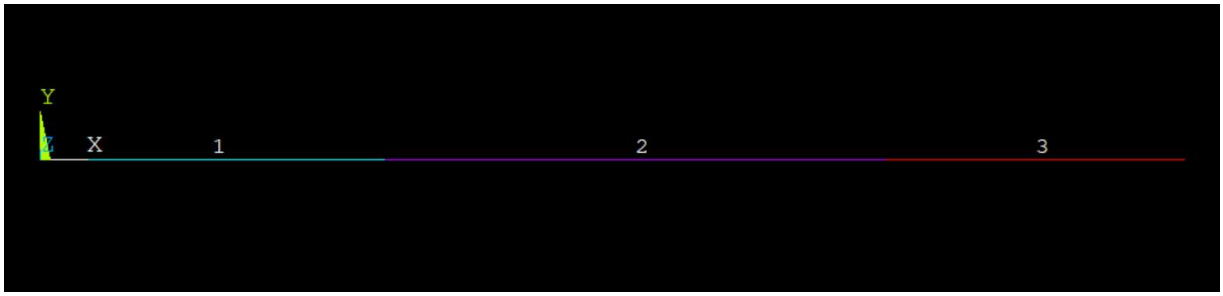
Table 2: *
Reaction forces and bending moment at the supports.

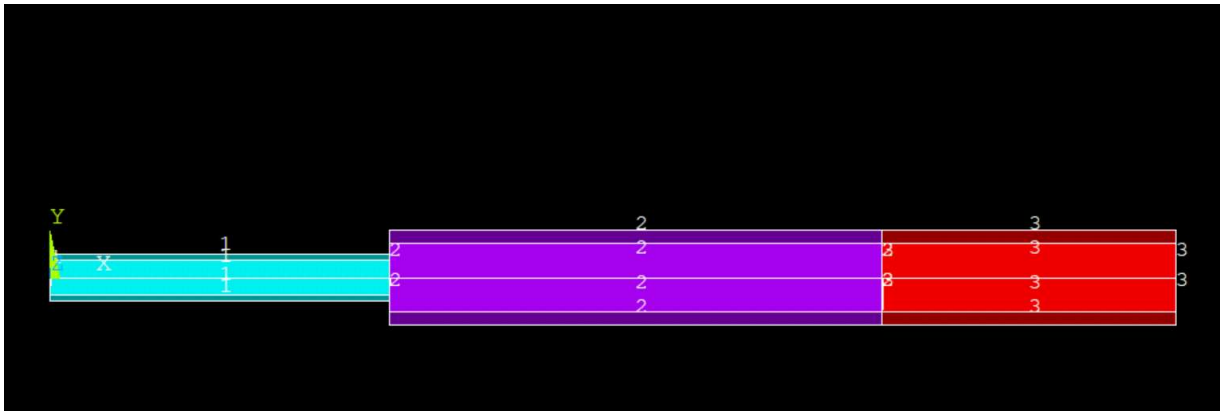
First Few Natural Frequencies

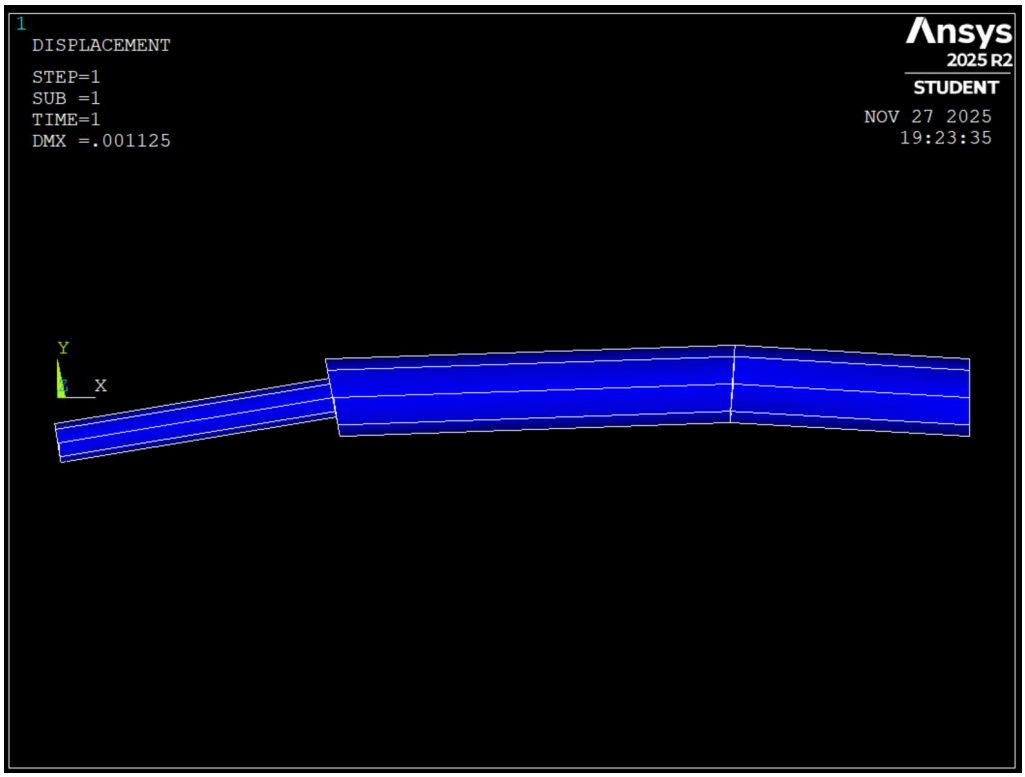
Mode	f_n [Hz]
1	1.143×10^{-3}
2	613.290
3	1663.800
4	2756.500
5	3391.800

Table 3: *
First few natural frequencies of the beam.

As Discussed during the consultation, these are the results agreed upon for the natural frequencies. while a different values are expected, it should have been within a close range with the values obtained from MATLAB


Y
X
1
2
3


Y
X
1
2
3


1
DISPLACEMENT
STEP=1
SUB =1
TIME=1
DMX =.001125
Ansys
2025 R2
STUDENT
NOV 27 2025
19:23:35
Y
X

PRNSOL Command

File

```
PRINT U    NODAL SOLUTION PER NODE

 ***** POST1 NODAL DEGREE OF FREEDOM LISTING *****

 LOAD STEP=     1  SUBSTEP=     1
  TIME=    1.0000      LOAD CASE=   0

 THE FOLLOWING DEGREE OF FREEDOM RESULTS ARE IN THE GLOBAL COORDINATE SYSTEM

   NODE       UY
      1 -0.11230E-002
      2   0.0000
      3  0.33938E-003
      4   0.0000

MAXIMUM ABSOLUTE VALUES
NODE          1
VALUE  -0.11230E-002
```

PRNSOL Command

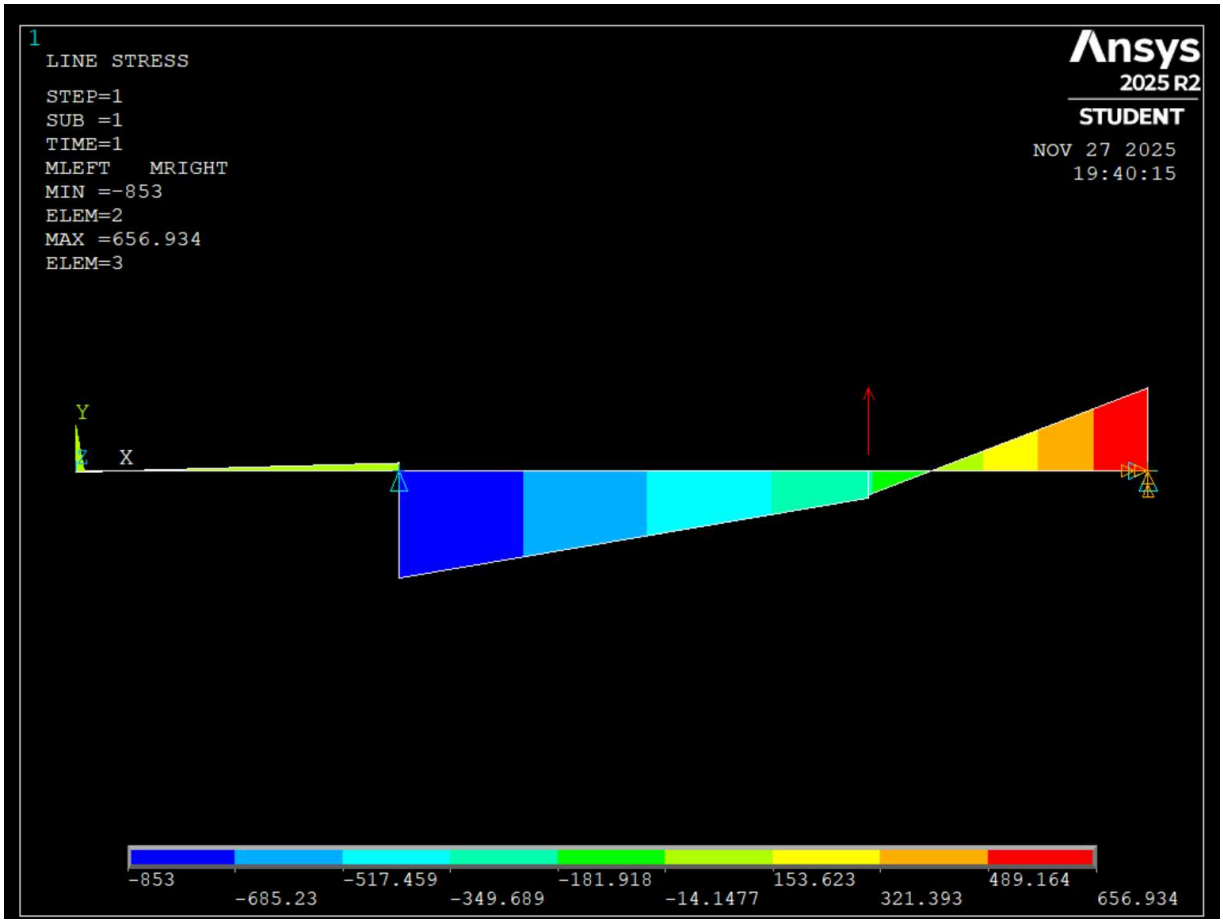
File

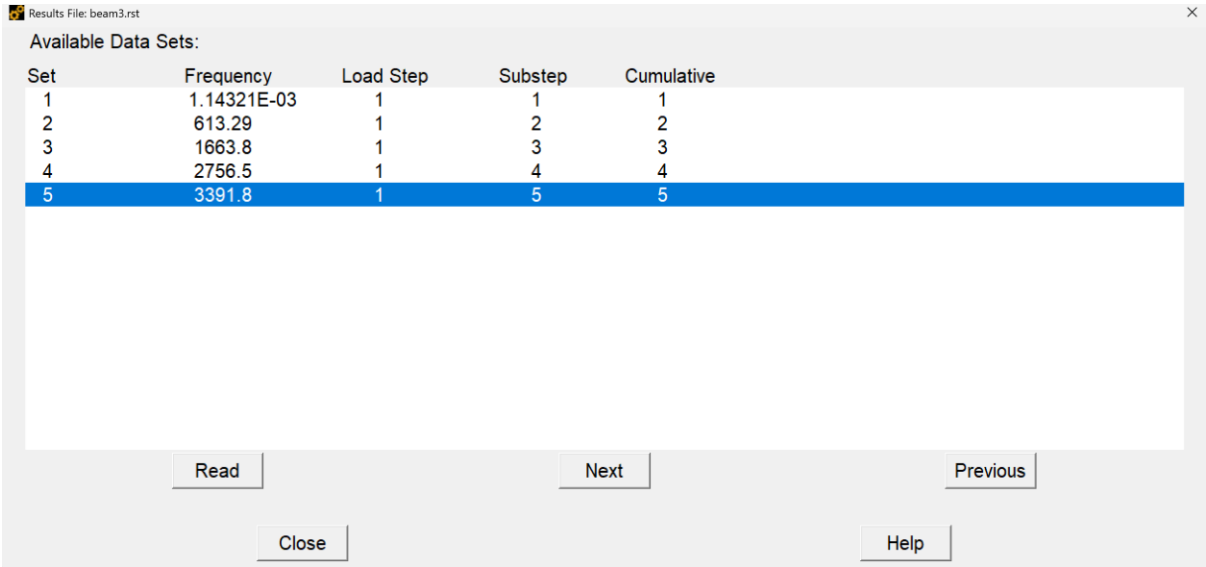
```
PRINT ROT  NODAL SOLUTION PER NODE

 ***** POST1 NODAL DEGREE OF FREEDOM LISTING *****

 LOAD STEP=     1  SUBSTEP=     1
  TIME=    1.0000      LOAD CASE=   0

 THE FOLLOWING DEGREE OF FREEDOM RESULTS ARE IN THE GLOBAL COORDINATE SYSTEM

   NODE       ROTZ
      1  0.48702E-002
      2  0.58509E-002
      3 -0.20353E-002
      4   0.0000

MAXIMUM ABSOLUTE VALUES
NODE          2
VALUE   0.58509E-002
```

PRRSOL Command

File

```
PRINT REACTION SOLUTIONS PER NODE

 ***** POST1 TOTAL REACTION SOLUTION LISTING *****

 LOAD STEP=     1  SUBSTEP=     1
  TIME=    1.0000      LOAD CASE=   0

 THE FOLLOWING X,Y,Z SOLUTIONS ARE IN THE GLOBAL COORDINATE SYSTEM

   NODE       FX          FY          FZ          MX          MY          MZ
      2                 846.54
      4   0.0000      -4466.5      0.0000      0.12524E-014-0.88130E-014  656.93

TOTAL VALUES
VALUE    0.0000      -3620.0      0.0000      0.12524E-014-0.88130E-014  656.93
```




Results File: beam3.rst
Available Data Sets:
Set Frequency Load Step Substep Cumulative
1 1.14321E-03 1 1 1
2 613.29 1 2 2
3 1663.8 1 3 3
4 2756.5 1 4 4
5 3391.8 1 5 5
Read
Next
Previous
Close
Help